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B. E. (Fifth Semester) Examination, April-May, 2016

(New Scheme)

(Elect. Engg. Branch)

COMMUNICATION THEORY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) in all questions is compulsory. Any two of three other sub parts of all questions are to be answered. Support the answer with neat diagams whenever necessary.

1. (a) Show the effect of different modulation indices on the waveform of AM signal. 2

(b) Explain the generation of DSB-SC signal using balanced modulator. 7

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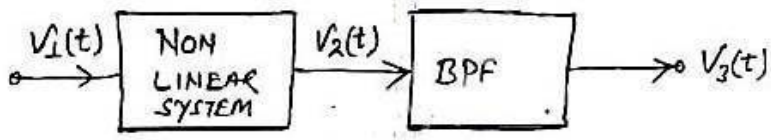
(c) Obtain the expression for the signal $V_3(t)$ in block diagram shown in fig. where

$$V_1(t) = 10 \cos 2000\,\pi t + 4 \cos 200\,\pi t \text{ and}$$

$$V_2(t) = V_1(t) + 0{\cdot}1\ V_1^2(t)$$

BPF filter is an ideal filter with passand 800th to 1200th. 7



(d) Explain demodulation of AM wave by using envelope detector. 7

2. (a) Explain the relationship between phase modulation and frequency modulation. 2

(b) The maximum deviation allowed in an FM broadcast system is 75 kHz. If the modulating signal is a single-tone sinusoid of 10 kHz, find the bandwidth of the FM signal. What will be the change in the bandwith, if modulating frequency is doubled? Determine the bandwidth when modulating signals amplitude is also doubled. 7

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(e) Compare the NBFM and WBFM. 7

(d) Explain Armstrong method of generating frequency modulation. 7

3. (a) State sampling theorem. 2

(b) Compare TDM and FDM. 7

(c) Explain PAM with the help of flat top sampling. 7

(d) How PWM signals are generated. 7

4. (a) Define ASK. 2

(b) A binary channel with bit rate $R_b = 36000$ b/s is available for PCM voice transmission. Find appropriate values of sampling rate f_s, the quantization level L and the binary digits n, assuming $f_m = 3{\cdot}2$ kHz. 7

(c) Explain Delta modulation. What is slope overloading in delta modulation? 7

(d) Explain the generation and reception of BPSK signal with suitable block diagram. 7

5. (a) What do you mean by information and its quality? 2

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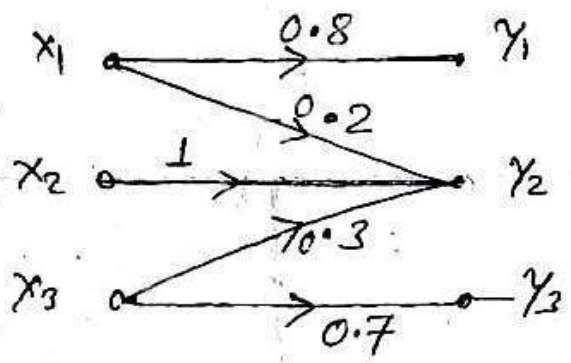
(b) An AWGN channel with 4 kHz bandwidth and noise power spectral density $\eta/2 = 10^{12}$ W/Hz. The signal power required at the receiver is 0·1 mW. Calculate the capacity of the channel. 7

(c) A discrets source transmit message x_1, x_2 and x_3 with prob-ability 0·3, 0.4 and 0·3. Calculate Mutual information. 7



(d) Explain the following : 7

(i) Source coding

(ii) The code length

(iii) Code efficiency

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